MATH 162, Winter 2026
SCRIPT 7: The Field Axioms

We will formalize the notions of addition and multiplication in structures called fields. A field with a compatible order is called an ordered field. We will see that $\mathbb{Q}$ and $\mathbb{R}$ are both examples of ordered fields.

Definition 7.1. A *binary operation* on a nonempty set X is a function

$$f\colon X \times X \longrightarrow X.$$

We say that f is *associative* if

$$f(f(x,y),z) = f(x,f(y,z)) \quad \text{for all } x,y,z \in X.$$

We say that f is *commutative* if

$$f(x,y) = f(y,x) \quad \text{for all } x,y \in X.$$

An *identity element* of a binary operation f is an element $e \in X$ such that

$$f(x,e) = f(e,x) = x \quad \text{for all } x \in X.$$

Remark 7.2. Frequently, we denote a binary operation differently. If $*\colon X \times X \longrightarrow X$ is the binary operation, we often write $a*b$ in place of $*(a,b)$. We sometimes indicate this same operation by writing $(a,b) \mapsto a*b$.

Exercise 7.3. Rewrite Definition 7.1 using the notation of Remark 7.2.

Exercise 7.3. Let X be a nonempty set. A *binary operation* on X is a function

$$*\colon X \times X \to X,$$

By **Remark 7.2** we write this as $(x,y) \mapsto x*y$.
We say that the binary operation $*$ is *associative* if

$$(x*y)*z = x*(y*z) \quad \text{for all } x,y,z \in X.$$

We say that $*$ is *commutative* if

$$x*y = y*x \quad \text{for all } x,y \in X.$$

An element $e \in X$ is called an *identity element* for $*$ if

$$x*e = e*x = x \quad \text{for all } x \in X.$$

□

Examples 7.4.

1. *The function* $+\colon \mathbb{Z} \times \mathbb{Z} \longrightarrow \mathbb{Z}$ *which sends a pair of integers* (m, n) *to* $+(m, n) = m + n$ *is a binary operation on the integers, called addition. Addition is associative, commutative and has identity element* 0*.*

2. *The maximum of* m *and* n*, denoted* $\max(m, n)$*, is an associative and commutative binary operation on* $\mathbb{Z}$*. Is there an identity element for* $\max$*?*

3. *Let* $\wp(Y)$ *be the power set of a set* Y*. Recall that the power set consists of all subsets of* Y*. Then the intersection of sets,* $(A, B) \mapsto A \cap B$*, defines an associative and commutative binary operation on* $\wp(Y)$*. Is there an identity element for* $\cap$*?*

Answer (Part (b)): For the maximum operation on $\mathbb{Z}$, the operation $\max(m, n)$ is associative and commutative because the maximum of two numbers does not depend on the order or grouping of the numbers. However, there is no identity element for max in $\mathbb{Z}$. An identity element e would have to satisfy $\max(m, e) = m$ for all $m \in \mathbb{Z}$, which would require $e \leq m$ for every integer m. Since the integers are unbounded below, no such e exists. Therefore, the maximum operation has no identity element in $\mathbb{Z}$.

Answer (Part (c)): For the intersection operation on $\wp(Y)$, the power set of a set Y, the operation $A \cap B$ is associative and commutative because the intersection of sets does not depend on order or grouping. There is an identity element, which is the set Y itself, since for any subset $A \subseteq Y$ we have $A \cap Y = A$. Therefore, Y serves as the identity element for the intersection operation.

Exercise 7.5. Find a binary operation on a set that is not commutative. Find a binary operation on a set that is not associative.

Proof. We want to provide two examples to show that a binary operation need not be commutative and need not be associative.

First, we want to provide an example of a binary operation that is not communicative. Let $X = \mathbb{Z}$, $m, n \in X$. Suppose we define a binary operation $*\colon X \times X \to X$ as

$$m * n = m - n$$

As the difference of two integers is again an integer, we may say this is a binary operation. However, $*$ is not commutative since in general $m - n \neq n - m$. For example, we may write,

$$5 * 3 = 2 \quad \text{while} \quad 3 * 5 = -2.$$

Thus, this work to prove that $m * n = m - n$ is a binary operation that is not commutative.

Next, let $Y = \mathbb{R}$ such that $x, y \in Y$. Suppose we define a binary operation $\circ\colon Y \times Y \to Y$ as

$$x \circ y = \frac{x + y}{2}.$$

We may consider this a binary operation as the average of two real numbers is again a real number. Now, consider $(x \circ y) \circ z$ and $x \circ (y \circ z)$:

$$(x \circ y) \circ z = \frac{\frac{x+y}{2} + z}{2} = \frac{x + y + 2z}{4},$$

while

$$x \circ (y \circ z) = \frac{x + \frac{y+z}{2}}{2} = \frac{2x + y + z}{4}.$$

Since these expressions are not equal in general, it fails to be associative. Therefore, there exist binary operations that are not commutative and binary operations that are not associative. $\square$

Exercise 7.6. Let X be a nonempty finite set, and let $Y = \{f : X \longrightarrow X \mid f \text{ is bijective}\}$. Consider the binary operation of composition of functions, denoted $\circ : Y \times Y \longrightarrow Y$ and defined by $(f \circ g)(x) = f(g(x))$, as seen in Definition 1.27. Decide whether or not composition is commutative and/or associative and whether or not it has an identity.

Exercise 7.6. DO NOT NEED TO PROVE $\square$

Theorem 7.7. *Identity elements are unique. That is, suppose that f is a binary operation on a set X that has two identity elements e and e'. Then $e = e'$.*

Proof of 7.7. Since e is an identity element for f, we have

$$f(e, e') = e'.$$

Since e' is also an identity element for f, we have

$$f(e, e') = e.$$

Therefore we can conclude that,

$$e = e'.$$

This works to prove that identity elements are unique. $\square$

Definition 7.8. A *field* is a set F with two binary operations on F called addition, denoted $+$, and multiplication, denoted $\cdot$, satisfying the following *field axioms*:

FA1 (Commutativity of Addition) For all $x, y \in F$, $x + y = y + x$.

FA2 (Associativity of Addition) For all $x, y, z \in F$, $(x+y)+z = x+(y+z)$.

FA3 (Additive Identity) There exists an element $0 \in F$ such that $x+0 = 0+x = x$ for all $x \in F$.

FA4 (Additive Inverses) For any $x \in F$, there exists $y \in F$ such that $x+y = y+x = 0$, called an additive inverse of x.

FA5 (Commutativity of Multiplication) For all $x, y \in F$, $x \cdot y = y \cdot x$.

FA6 (Associativity of Multiplication) For all $x, y, z \in F$, $(x \cdot y) \cdot z = x \cdot (y \cdot z)$.

FA7 (Multiplicative Identity) There exists an element $1 \in F$ such that $x \cdot 1 = 1 \cdot x = x$ for all $x \in F$.

FA8 (Multiplicative Inverses) For any $x \in F$ such that $x \neq 0$, there exists $y \in F$ such that $x \cdot y = y \cdot x = 1$, called a multiplicative inverse of x.

FA9 (Distributivity of Multiplication over Addition) For all $x, y, z \in F$, $x \cdot (y+z) = x \cdot y + x \cdot z$.

FA10 (Distinct Additive and Multiplicative Identities) $1 \neq 0$.

Exercise 7.9. Consider the set $\mathbb{F}_2 = \{0, 1\}$, and define binary operations $+$ and $\cdot$ on $\mathbb{F}_2$ by:

$$\begin{array}{cccc} 0+0=0 & 0+1=1 & 1+0=1 & 1+1=0 \\ 0 \cdot 0 = 0 & 0 \cdot 1 = 0 & 1 \cdot 0 = 0 & 1 \cdot 1 = 1 \end{array}$$

Show that $\mathbb{F}_2$ is a field.

Exercise 7.9. To verify that $\mathbb{F}_2$ satisfies FA1, it suffices to check all possible sums in $\mathbb{F}_2$. In particular, we verify that

$$0+0 = 0+0, \quad 0+1 = 1+0, \quad \text{and} \quad 1+1 = 1+1.$$

The first and third equalities are immediate. For the second, we have $0+1 = 1 = 1+0$, so FA1 holds. To verify that $\mathbb{F}_2$ satisfies FA2, it suffices to proceed by casework.

$$\begin{array}{ll} (0+0)+0 = 0 = 0+(0+0) & (0+0)+1 = 1 = 0+(0+1) \\ (0+1)+0 = 1 = 0+(1+0) & (1+0)+0 = 1 = 1+(0+0) \\ (0+1)+1 = 0 = 0+(1+1) & (1+1)+0 = 0 = 1+(1+0) \\ (1+0)+1 = 0 = 1+(0+1) & (1+1)+1 = 1 = 1+(1+1) \end{array}$$

To verify that $\mathbb{F}_2$ satisfies FA3, it suffices to show that there exists an element $0 \in \mathbb{F}_2$ such that $x+0 = 0+x = x$ for all $x \in \mathbb{F}_2$. We observe that $0+0=0$ and $1+0=1$. By commutativity of addition, we also have $0+1=1$. Therefore, 0 acts as an additive identity in $\mathbb{F}_2$.

To show that $\mathbb{F}_2$ satisfies FA4, it suffices to verify that for every $x \in \mathbb{F}_2$, there exists an element $y \in \mathbb{F}_2$ such that $x+y = y+x = 0$. When $x=0$, we may take $y=0$, since $0+0=0$. When $x=1$, we may take $y=1$, since $1+1=0$. Hence, every element of $\mathbb{F}_2$ has an additive inverse.

To show that $\mathbb{F}_2$ satisfies FA5, it suffices to verify that $0 \cdot 0 = 0 \cdot 0$, $0 \cdot 1 = 1 \cdot 0$, and $1 \cdot 1 = 1 \cdot 1$. $0 \cdot 1 = 0 = 1 \cdot 0$, and hence the condition is satisfied. The first and third equalities are immediate. For the second, we observe that $0 \cdot 1 = 0 = 1 \cdot 0$, and hence the condition is satisfied.

To prove that $\mathbb{F}_2$ obeys FA6, the following casework will suffice.

$$\begin{aligned}
(0 \cdot 0) \cdot 0 &= 0 = 0 \cdot (0 \cdot 0) & (0 \cdot 0) \cdot 1 &= 0 = 0 \cdot (0 \cdot 1) \\
(0 \cdot 1) \cdot 0 &= 0 = 0 \cdot (1 \cdot 0) & (1 \cdot 0) \cdot 0 &= 0 = 1 \cdot (0 \cdot 0) \\
(0 \cdot 1) \cdot 1 &= 0 = 0 \cdot (1 \cdot 1) & (1 \cdot 1) \cdot 0 &= 0 = 1 \cdot (1 \cdot 0) \\
(1 \cdot 0) \cdot 1 &= 0 = 1 \cdot (0 \cdot 1) & (1 \cdot 1) \cdot 1 &= 1 = 1 \cdot (1 \cdot 1)
\end{aligned}$$

To show that $\mathbb{F}_2$ satisfies FA7, it suffices to identify an element $1 \in \mathbb{F}_2$ such that $x \cdot 1 = 1 \cdot x = x$ for all $x \in \mathbb{F}_2$. Since $0 \cdot 1 = 0$ and $1 \cdot 1 = 1$, and using commutativity, it follows that 1 acts as a multiplicative identity in $\mathbb{F}_2$.

To show that $\mathbb{F}_2$ satisfies FA8, it suffices to verify that for every $x \in \mathbb{F}_2$ with $x \neq 0$, there exists a $y \in \mathbb{F}_2$ such that $x \cdot y = y \cdot x = 1$. In the case $x = 1$, this element is 1, since $1 \cdot 1 = 1 \cdot 1 = 1$.
To prove that $\mathbb{F}_2$ obeys FA9, the following casework will suffice.

$$\begin{aligned}
0 \cdot (0 + 0) &= 0 = 0 \cdot 0 + 0 \cdot 0 & 0 \cdot (0 + 1) &= 0 = 0 \cdot 0 + 0 \cdot 1 \\
0 \cdot (1 + 0) &= 0 = 0 \cdot 1 + 0 \cdot 0 & 1 \cdot (0 + 0) &= 0 = 1 \cdot 0 + 1 \cdot 0 \\
0 \cdot (1 + 1) &= 0 = 0 \cdot 1 + 0 \cdot 1 & 1 \cdot (1 + 0) &= 1 = 1 \cdot 1 + 1 \cdot 0 \\
1 \cdot (0 + 1) &= 1 = 1 \cdot 0 + 1 \cdot 1 & 1 \cdot (1 + 1) &= 0 = 1 \cdot 1 + 1 \cdot 1
\end{aligned}$$

To prove that $\mathbb{F}_2$ obeys FA10, it will suffice to show that $0 \neq 1$. Clearly this is true.
This completes the proof. $\square$

Theorem 7.10. *Suppose that F is a field. Then additive inverses are unique. This means: Let $x \in F$. If $y, y' \in F$ satisfy $x + y = 0$ and $x + y' = 0$, then $y = y'$.*

Final Proof of 7.10 - to check. Let $x \in F$. If $y, y' \in F$, such taht $x + y = 0$ and $x + y' = 0$ then we want to show that $y = y'$.

By the Additive Identity (FA3), we may state that

$$y = y + 0$$

Then, as $x + y' = 0$, we may say that

$$y = y + 0 = y + (x + y')$$

We may write this as

$$y = y + (x + y') = (y + x) + y'$$

As $(x + y) = 0$, we can rewrite this as

$$y = (y + x) + y' = 0 + y' = y'$$

$$y = y'$$

As $y = y'$, this works to prove that additive inverses are unique.

$\square$

We usually write $-x$ for the additive inverse of x.

Corollary 7.11. *If $x \in F$, then $-(-x) = x$.*

Final Proof of Corollary 7.11- need to check. Let $x \in F$. We want to prove that $-(-x) = x$.

By the definition of the additive inverse, we may say that $x + (-x) = 0$. Then, let us take the additive inverse of both sides. This would get us:

$$-(x + (-x)) = -(0)$$

By the law of distribution, we may write this as

$$(-x) + -(-x) = 0$$

By adding the two equations together, we may state that

$$x + (-x) = (-x) + -(-x)$$

By subtracting $(-x)$ from both sides, we get that

$$x = -(-x)$$

This works to prove that $x = -(-x)$. $\square$

Theorem 7.12. *Let F be a field, and let $a, b, c \in F$. If $a + b = a + c$, then $b = c$.*

Proof of 7.12. Let F be a field and let $a, b, c, \in F$. If $a + b = a + c$, we want to show that $b = c$.

Suppose $\exists(-a) \in F$ such that $a + (-a) = 0$ (FA4). Now, let the additive inverse $-a$ be added on both sides of the equations such that

$$a + b + (-a) = a + c + (-a)$$

We may rewrite the equation as

$$(a + -a) + b = (a + -a) + c$$

Therefore, from this we get that

$$0 + b = 0 + c \text{ by FA3}$$

$$b = c \text{ by FA3}$$

Therefore, this works to prove that $b = c$. ☐

Theorem 7.13. *Let F be a field. If $a \in F$, then $a \cdot 0 = 0$.*

Final Proof of Theorem 7.13. Let F be a field and $a \in F$. We want to prove that $a \cdot 0 = 0$.

Suppose there exists some $x \in F$. Firstly, we want to show that if $x + x = x$, then $x = 0$.

Consider the equation $x + x = x$. If we add the additive inverse -x onto both sides we get the equation

$$(-x) + (x + x) = (-x) + x$$

We can rewrite the equation as

$$(-x + x) + x = (-x + x)$$

By the field axion FA4, we may rewrite this as

$$x = 0$$

Hence, we get that $x = 0$.

Now consider $a \cdot 0$. Using distributivity (FA9), we have

$$a \cdot (0 + 0) = a \cdot 0 + a \cdot 0.$$

But $0 + 0 = 0$, so the left-hand side simplifies to

$$a \cdot 0 = a \cdot 0 + a \cdot 0.$$

Setting $x = a \cdot 0$, we see that $x + x = x$, and by the result above, it follows that

$$a \cdot 0 = 0.$$

This completes the proof. ☐

Theorem 7.14. *Suppose that F is a field. Then multiplicative inverses are unique. This means: Let $x \in F$. If $y, y' \in F$ satisfy $x \cdot y = 1$ and $x \cdot y' = 1$, then $y = y'$.*

Proof of Theorem 7.14. Suppose there exists a field F. Let $x \in F$. We want to show that if $y, y' \in F$ satisfy $x \cdot y = 1$ and $x \cdot y' = 1$ then $y = y'$.

As $x \cdot y = 1$ and $x \cdot y' = 1$, consider the following equation

$$x \cdot y = x \cdot y'$$

Now, we can further multiply both sides of this equation by another multiplicative inverse $z \in F$ of x. Therefore, we may write this expression as

$$z \cdot (x \cdot y) = z \cdot (x \cdot y')$$

We can rewrite this equation as

$$(z \cdot x) \cdot y = (z \cdot x) \cdot y'$$

Then, by FA8, we may write $z \cdot x = 1$. Therefore, we can simplify the equation as

$$1 \cdot y = 1 \cdot y'$$

By the multiplicative identity, we may further simplify the equation to

$$y = y'$$

Thus, this works to prove that $y = y'$. $\square$

We usually write x^{-1} or $\frac{1}{x}$ for the multiplicative inverse of x.

Corollary 7.15. *Note that you need 7.13 to justify the existence of $(x^{-1})^{-1}$.*
If $x \in F$ and $x \neq 0$, then $(x^{-1})^{-1} = x$.

Proof of Corollary 7.15. Suppose there exists a field F. Let $x \in F$ such that $x \neq 0$. We want to prove that $(x^{-1})^{-1} = x$.

Since $x \neq 0$, by FA8 there exists a multiplicative inverse $x^{-1} \in F$ such that

$$x \cdot x^{-1} = 1.$$

Furthermore, $x^{-1} \neq 0$, because if $x^{-1} = 0$, then

$$x \cdot x^{-1} = x \cdot 0 = 0 \neq 1,$$

which contradicts the definition of x^{-1}. By FA8 applied to x^{-1}, there exists $y \in F$ such that

$$y \cdot x^{-1} = x^{-1} \cdot y = 1.$$

By definition, $y = (x^{-1})^{-1}$.

Now, multiply both sides of $y \cdot x^{-1} = 1$ on the right by x:

$$(y \cdot x^{-1}) \cdot x = 1 \cdot x.$$

By associativity of multiplication (FA6) and the fact that $x^{-1} \cdot x = 1$, we have

$$y \cdot (x^{-1} \cdot x) = y \cdot 1 = y = x.$$

Hence,

$$(x^{-1})^{-1} = y = x,$$

Now, consider the equation

$$(x^{-1})^{-1} = x.$$

By the existence of multiplicative inverses in a field, and since $x^{-1} \neq 0$, there exists a multiplicative inverse of x^{-1}, which we denote by $(x^{-1})^{-1}$.
By definition of a multiplicative inverse, we have

$$x^{-1} \cdot (x^{-1})^{-1} = 1 \quad \text{and} \quad x \cdot x^{-1} = 1.$$

Since both products equal 1, it follows that

$$x^{-1} \cdot (x^{-1})^{-1} = x \cdot x^{-1}.$$

Multiplying both sides on the right by x and by FA6, we obtain

$$(x^{-1} \cdot (x^{-1})^{-1}) \cdot x = (x \cdot x^{-1}) \cdot x \implies (x^{-1})^{-1} = x.$$

Therefore, the multiplicative inverse of the inverse of x is indeed x. $\square$

Theorem 7.16. *Let F be a field, and let $a, b, c \in F$. If $a \cdot b = a \cdot c$ and $a \neq 0$, then $b = c$.*

Proof of Theorem 7.16. Let there be a field F. Then, let $a, b, c \in F$ such that $a \cdot b = a \cdot c$ where $a \neq 0$. We want to show that $b = c$.

Let $a^{-1} \in F$. As $a \neq 0$, by FA8, we may say that $a \cdot a^{-1} = 1$. Now, consider the equation $a \cdot b = a \cdot c$. We may multiply both sides by the multiplicative inverse to get

$$a^{-1} \cdot a \cdot b = a^{-1} \cdot a \cdot c$$
$$(a^{-1} \cdot a) \cdot b = (a^{-1} \cdot a) \cdot c$$
$$1 \cdot b = 1 \cdot c$$

By the field axiom FA7, we may state $1 \cdot b = b$ and $1 \cdot c = c$. Therefore, we get that $b = c$.

Thus, this works to prove that if $a \cdot b = a \cdot c$ and $a \neq 0$, then $b = c$. $\square$

Theorem 7.17. *Let F be a field, and let $a, b \in F$. If $a \cdot b = 0$, then $a = 0$ or $b = 0$.*

Proof of 7.17. Let F be a field, and let $a, b \in F$. We want to prove that if $a \cdot b = 0$, then $a = 0$ or $b = 0$.

Assume that $a \neq 0$. Since F is a field, a has a multiplicative inverse a^{-1}. Then, we may multiply both sides of the equation $a \cdot b = 0$ by a^{-1} to get the equation

$$a^{-1}(a \cdot b) = a^{-1} \cdot 0.$$

By FA6 we can state that,

$$(a^{-1}a) \cdot b = 0.$$

Since $a^{-1}a = 1$ and $1 \cdot b = b$, this gives

$$b = 0.$$

Therefore, if $a \cdot b = 0$, then either $a = 0$ or $b = 0$. □

Lemma 7.18. *Let F be a field. If $a \in F$, then $-a = (-1)a$.*

Proof. Since 0 is the additive identity, we have

$$0 = 0 \cdot a.$$

Using the fact that $0 = (-1) + 1$ in a field, we may write

$$0 \cdot a = ((-1) + 1) \cdot a.$$

By the distributive property,

$$((-1) + 1) \cdot a = (-1) \cdot a + 1 \cdot a.$$

Since $1 \cdot a = a$, this becomes

$$0 = (-1) \cdot a + a.$$

Thus $(-1) \cdot a$ is the additive inverse of a, and by uniqueness of additive inverses,

$$(-1) \cdot a = -a.$$

□

Lemma 7.19. *Let F be a field. If $a, b \in F$, then*

$$a \cdot (-b) = -(a \cdot b) = (-a) \cdot b.$$

Proof of Lemma 7.19. Let $a, b \in F$.

We first show that $a \cdot (-b) = -(a \cdot b)$. Since $-b$ is the additive inverse of b, we have

$$b + (-b) = 0.$$

Therefore, if we plug it back into the equation, we have that

$$a \cdot (b + (-b)) = a \cdot 0.$$

By distributivity and the fact that $a \cdot 0 = 0$, this becomes

$$a \cdot b + a \cdot (-b) = 0.$$

Thus, $a \cdot (-b)$ is the additive inverse of $a \cdot b$, which means

$$a \cdot (-b) = -(a \cdot b).$$

This works to prove that $a \cdot (-b) = -(a \cdot b)$.

Next, we show that $(-a) \cdot b = -(a \cdot b)$. Since $-a$ is the additive inverse of a, we have

$$a + (-a) = 0.$$

Multiplying both sides by b and applying distributivity yields

$$(a + (-a)) \cdot b = 0 \cdot b.$$

Again using distributivity and the fact that $0 \cdot b = 0$, we obtain

$$a \cdot b + (-a) \cdot b = 0.$$

This shows that $(-a) \cdot b$ is also the additive inverse of $a \cdot b$, so

$$(-a) \cdot b = -(a \cdot b).$$

From the two results, we may therefore conclude that

$$a \cdot (-b) = -(a \cdot b) = (-a) \cdot b.$$

□

Lemma 7.20. *Let F be a field. If $a, b \in F$, then $a \cdot b = (-a) \cdot (-b)$.*

Proof of Lemma 7.20. Let $a, b \in F$. By **Lemma 7.19**, we know that

$$(-a) \cdot b = -(a \cdot b) \quad \text{and} \quad a \cdot (-b) = -(a \cdot b)$$

Now, consider the expression $(-a) \cdot (-b)$. By the fact that $(-a) \cdot b = -(a \cdot b)$, we may write this expression as

$$(-a) \cdot (-b) = -\big((-a) \cdot b\big).$$

Furthermore, we may perform the same operation using the principle that $a \cdot (-b) = -(a \cdot b)$. Therefore, we may write

$$(-a) \cdot (-b) = -\big((-a) \cdot (-b)\big) = -(-(a \cdot b))$$

By **Corollary 7.11**, we know that means that taking the negative of a negative returns the original value. Therefore, we get that:

$$-(-(a \cdot b)) = a \cdot b.$$

Therefore, we have that

$$(-a) \cdot (-b) = a \cdot b,$$

This works to prove that $a \cdot b = (-a) \cdot (-b)$. $\square$

Next, we discuss the notion of an ordered field.

Definition 7.21. An *ordered field* is a field F equipped with an ordering $<$ (satisfying Definition 3.1) such that also:

- Addition respects the ordering: if $x < y$, then $x + z < y + z$ for all $z \in F$.
- Multiplication respects the ordering: if $0 < x$ and $0 < y$, then $0 < x \cdot y$.

Definition 7.22. Suppose F is an ordered field and $x \in F$. If $0 < x$, we say that x is *positive*. If $x < 0$, we say that x is *negative*.

For the remaining theorems, let F be an ordered field.

Lemma 7.23. *If $0 < x$, then $-x < 0$. Similarly, if $x < 0$, then $0 < -x$.*

Proof of Lemma 7.23. Suppose first that $0 < x$. By the order axioms of an ordered field, adding the same element to both sides of an inequality preserves the inequality. Adding $-x$ to both sides of $0 < x$, we obtain

$$0 + (-x) < x + (-x).$$

Simplifying both sides gives

$$-x < 0.$$

This works to prove that if $0 < x$ then $-x < 0$.
Now suppose that $x < 0$. Again, adding $-x$ to both sides of the inequality yields

$$x + (-x) < 0 + (-x).$$

Simplifying gives

$$0 < -x.$$

This works to prove that if $x < 0$ then $0 < -x$. This completes the proof. $\square$

Lemma 7.24. *Let $x, y, z \in F$.*

1. *If $x > 0$ and $y < z$, then $x \cdot y < x \cdot z$.*
2. *If $x < 0$ and $y < z$, then $x \cdot z < x \cdot y$.*

Proof of Lemma 7.24. **Part (1):** Suppose $x > 0$ and $y < z$.

First, we can substract y from both sides of the inequality $y < z$ gives

$$y + (-y) < z + (-y)$$

$$z - y > 0.$$

Then, we can multiply both sides by x, which is positive, and thereby yields

$$x \cdot (z - y) > 0.$$

Then, using distributivity, we can expand $x \cdot (z - y)$ to get:

$$x \cdot (z - y) = x \cdot z - x \cdot y.$$

By substituting this into the inequality, we obtain

$$x \cdot z - x \cdot y > 0.$$

Then, adding $x \cdot y$ to both sides gives

$$x \cdot z - x \cdot y + (x \cdot y) > 0 + (x \cdot y)$$

$$x \cdot z > x \cdot y.$$

This works to prove that if $x > 0$ and $y < z$, then $x \cdot y < x \cdot z$.

Part (2): Suppose $x < 0$ and $y < z$.

As before, $z - y > 0$. Since $x < 0$, we have $-x > 0$. Multiplying $z - y$ by $-x$ gives

$$(-x) \cdot (z - y) > 0.$$

Expanding using distributivity, we get

$$(-x) \cdot (z - y) = (-x) \cdot z - (-x) \cdot y.$$

By the lemma on negatives, $(-x) \cdot z = -(x \cdot z)$ and $(-x) \cdot y = -(x \cdot y)$. Substituting, we obtain

$$-(x \cdot z) - (-(x \cdot y)) > 0 \quad \Rightarrow \quad -(x \cdot z) + x \cdot y > 0.$$

Rearranging terms gives

$$x \cdot y - x \cdot z > 0$$

$$x \cdot z < x \cdot y.$$

This works to prove that if $x < 0$ and $y < z$, then $x \cdot z < x \cdot y$. □

Remark 7.25. An immediate consequence of this lemma is the fact that if x and y are both positive or both negative, their product is positive.

Lemma 7.26. *If $x \in F$, then $0 \leq x^2$. Moreover, if $x \neq 0$, then $0 < x^2$.*

Proof of Lemma 7.26. Firstly, if $x \in F$, we want to show that $0 \leq x^2$. We will do this through three cases:

Case 1: $x = 0$. Let $x \in F$ such that $x = 0$. Therefore we may write $0 \leq (x \cdot x)$ as $0 \leq 0 \cdot 0$. Then, we get the inequality, $0 \leq 0$ which we know to be true.

Case 2: $x > 0$. Let $x \in F$ such that $x > 0$. Therefore we may write $0 \leq x^2$ as $0 \leq (x \cdot x)$. As $x > 0$, we know that $x \cdot x > 0$. Therefore, have that $0 \leq (x \cdot x)$ stands true.

Case 3: $x < 0$. Let $x \in F$ such that $x < 0$. By the properties of additive inverses, we have $-x > 0$. Applying the result from Case 2 to the positive element $(-x)$, we know that $(-x) \cdot (-x) > 0$. By **Lemma 7.20**, we know that $(-x) \cdot (-x) = x \cdot x = x^2$. Therefore, $x^2 > 0$, and the inequality $0 \leq x^2$ holds.

This works to prove that if $x \in F$ then $0 \leq x^2$. Now, to show that if $x \neq 0$ then $0 < x^2$ consider only Case 2 and Case 3.

Case 2: We see that if $x > 0$ we have that $x \cdot x > 0$ as $x \neq 0$. Therefore, as $x \cdot x > 0$ we have that $x^2 > 0$ and therefore, we may say that $0 < x^2$.

Case 3: We see that if $x < 0$ then we have that $(-x) \cdot (-x) > 0$. Therefore, from here we get that if $x \neq 0$ then $x^2 > 0$.

Therefore, this works to prove that if $x \neq 0$ then we have that $0 < x^2$. □

Corollary 7.27. $0 < 1$.

Proof of Corollary 7.27. We want to prove that $0 < 1$. By **Lemma 7.26**, we know that $0 \leq x^2$.

Let $1 \in F$. Therefore, we know that, by **Field Axiom 10 (FA10)**, there exists a multiplicative inverse y such that $1 \cdot y = 1$. By **Field Axiom 7 (FA7)**, we know that for all $y \in F$ we may write that $1 \cdot y = y$. As both axioms must be true, we get that the multiplicative inverse of 1 is 1 and that $1 \cdot 1 = 1$.

Now, consider **Lemma 7.26**. We have that $0 \leq x^2$. Let $x = 1$. Therefore, we have that $0 \leq 1 \cdot 1$. We may write this as $0 \leq 1$. As we know that identity elements are unique, **Theorem 7.7**, we know that $0 \neq 1$. This works to prove that $0 < 1$. □

Proof of Theorem 7.28. Let F be an ordered field.

Suppose for the sake of contradiction that F has a last point, $a \in F$. By definition of a last point, this means that for all $x \in F$, $x \leq a$. Now, consider the element $1 \in F$. From **Corollary 7.27**, we know that $0 < 1$. By the additive property of ordered fields, we can add a to both sides of this inequality:

$$0 + a < 1 + a \implies a < a + 1$$

By the closure of addition in a field, $a + 1$ is an element of F. However, we now have an element $a + 1 \in F$ that is strictly greater than a. As a is an arbitrary element of the ordered field, we may say that this applies for all elements of the ordered field. This contradicts the assumption that a is the last point. Thus, F has no last point.

To show that F has no first point, we use analogous reasoning. Suppose there exists a first point $b \in F$, meaning $b \leq x$ for all $x \in F$. Since $0 < 1$, by the properties of additive inverses, we know that $-1 < 0$. Adding b to both sides gives:

$$b + (-1) < b + 0 \implies b - 1 < b$$

Since $b - 1 \in F$ and $b - 1 < b$, and that b is an arbitrary element, this contradicts the assumption that b is the first point. Thus, F has no first point. □

Theorem 7.28. *The rational numbers $\mathbb{Q}$ form an ordered field.*

Proof. Let $\left[\frac{a}{b}\right], \left[\frac{c}{d}\right], \left[\frac{x}{y}\right] \in \mathbb{Q}$ with $b, d, y > 0$, and suppose that

$$\left[\tfrac{a}{b}\right] < \left[\tfrac{c}{d}\right].$$

By definition of the order on $\mathbb{Q}$, this means that $ad < bc$. Since $y > 0$, we may multiply both sides of this inequality by y^2 without changing its direction, obtaining

$$ad\, y^2 < bc\, y^2.$$

Adding the same quantity $bdxy$ to both sides preserves the inequality, so we have

$$ad\, y^2 + bdxy < bc\, y^2 + bdxy.$$

Rearranging terms yields

$$(ay + bx)(dy) < (by)(cy + dx).$$

Because $by > 0$ and $dy > 0$, this inequality is equivalent to

$$\left[\tfrac{ay+bx}{by}\right] < \left[\tfrac{cy+dx}{dy}\right].$$

By the definition of addition in $\mathbb{Q}$, we have

$$\left[\tfrac{ay+bx}{by}\right] = \left[\tfrac{a}{b}\right] + \left[\tfrac{x}{y}\right] \quad \text{and} \quad \left[\tfrac{cy+dx}{dy}\right] = \left[\tfrac{c}{d}\right] + \left[\tfrac{x}{y}\right].$$

Therefore,
$$\left[\tfrac{a}{b}\right] + \left[\tfrac{x}{y}\right] < \left[\tfrac{c}{d}\right] + \left[\tfrac{x}{y}\right],$$
which shows that addition in $\mathbb{Q}$ respects the ordering.

Secondly, we wnat to show that multiplication respects the ordering when multiplying positive rational numbers. Suppose $[\frac{a}{b}] < [\frac{c}{d}]$ and $[\frac{0}{1}] < [\frac{x}{y}]$ with $b, d, y > 0$. We can multiply the inequality $ad < bc$ by the positive integer xy:
$$(ad)(xy) < (bc)(xy)$$
By commutativity of integers, we rearrange this to:
$$(ax)(dy) < (by)(cx)$$
Since $by > 0$ and $dy > 0$, this final inequality is exactly the definition for:
$$\left[\tfrac{ax}{by}\right] < \left[\tfrac{cx}{dy}\right]$$
By the definition of multiplication in $\mathbb{Q}$:
$$\left[\tfrac{a}{b}\right] \cdot \left[\tfrac{x}{y}\right] < \left[\tfrac{c}{d}\right] \cdot \left[\tfrac{x}{y}\right]$$
This confirms the multiplicative property and thereby completes the proof.

Theorem 7.29. *Any ordered field F contains a copy of the rational numbers $\mathbb{Q}$ in the following sense. There exists an injective map $i\colon \mathbb{Q} \longrightarrow F$ that respects all of the axioms for an ordered field. In particular:*

- $i(0_{\mathbb{Q}}) = 0_F$.
- $i(1_{\mathbb{Q}}) = 1_F$.
- *If $a, b \in \mathbb{Q}$, then $i(a + b) = i(a) + i(b)$.*
- *If $a, b \in \mathbb{Q}$, then $i(a \cdot b) = i(a) \cdot i(b)$.*
- *If $a, b \in \mathbb{Q}$ and $a < b$, then $i(a) < i(b)$.*

Hint: start by defining a function $j\colon \mathbb{Z} \longrightarrow F$ satisfying the properties above, then use j to define i.

Proof of Theorem 7.29. Define $j\colon \mathbb{Z} \longrightarrow F$ inductively by $j(0) = 0$, $j(n) = j(n-1) + 1$ for $n \in \mathbb{N}$, and $j(-n) = -j(n)$. We first verify that j respects the structure of $\mathbb{Z}$.

1. $j(0) = 0$ and $j(1) = 1$ follow by definition.
2. To show $j(a + b) = j(a) + j(b)$, we consider the case where $a, b \geq 0$ and use induction on a. For $a = 0$, $j(0 + b) = j(b) = 0 + j(b) = j(0) + j(b)$. For $a \geq 1$, we have $j(a + b) = j((a - 1 + b) + 1) = j(a - 1 + b) + 1$. By the inductive hypothesis, this is $j(a-1) + j(b) + 1 = (j(a - 1) + 1) + j(b) = j(a) + j(b)$. The cases involving negative integers follow similarly by using the property $j(-n) = -j(n)$ and the additive axioms of F.

3. To show $j(a \cdot b) = j(a) \cdot j(b)$, we first consider $a, b \geq 0$ and use induction on a. For $a = 0$, $j(0 \cdot b) = j(0) = 0 = j(0) \cdot j(b)$. For $a \geq 1$, we have $j(a \cdot b) = j((a-1)b + b) = j((a-1)b) + j(b)$. By the inductive hypothesis, this is $j(a-1)j(b) + j(b) = (j(a-1)+1)j(b) = j(a)j(b)$. If $a < 0$ and $b < 0$, then $a \cdot b = (-a) \cdot (-b)$. Since $-a, -b > 0$, we have $j(ab) = j(-a)j(-b) = (-j(a))(-j(b)) = j(a)j(b)$.

4. To show j is order-preserving, let $a < b$ in $\mathbb{Z}$. Then $b - a = n$ for some $n \in \mathbb{N}$. We show $j(n) > 0$ by induction. For $n = 1$, $j(1) = 1 > 0$ by Corollary 7.27. For $n \geq 2$, $j(n) = j(n-1) + 1$. Since $j(n-1) > 0$ and $1 > 0$, $j(n) > 0$. Thus $j(b) - j(a) = j(b-a) > 0$, so $j(a) < j(b)$.

Now, we define $i\colon \mathbb{Q} \longrightarrow F$ by $i(\frac{a}{b}) = j(a) \cdot [j(b)]^{-1}$. We assume without loss of generality that $b > 0$.

1. We first show i is well-defined. Suppose $\frac{a}{b} = \frac{c}{d}$. Then $ad = bc$ in $\mathbb{Z}$. Applying j, we have $j(ad) = j(bc)$, which implies $j(a)j(d) = j(b)j(c)$. Since $b, d \neq 0$ and j is order-preserving, $j(b), j(d) \neq 0$. Dividing by $j(b)j(d)$ yields $\frac{j(a)}{j(b)} = \frac{j(c)}{j(d)}$, thus $i(\frac{a}{b}) = i(\frac{c}{d})$.

2. $i(0) = \frac{j(0)}{j(1)} = 0$ and $i(1) = \frac{j(1)}{j(1)} = 1$.

3. $i(\frac{a}{b} + \frac{c}{d}) = i(\frac{ad+bc}{bd}) = \frac{j(ad+bc)}{j(bd)} = \frac{j(a)j(d)+j(b)j(c)}{j(b)j(d)} = \frac{j(a)}{j(b)} + \frac{j(c)}{j(d)} = i(\frac{a}{b}) + i(\frac{c}{d})$.

4. $i(\frac{a}{b} \cdot \frac{c}{d}) = i(\frac{ac}{bd}) = \frac{j(ac)}{j(bd)} = \frac{j(a)j(c)}{j(b)j(d)} = \frac{j(a)}{j(b)} \cdot \frac{j(c)}{j(d)} = i(\frac{a}{b}) \cdot i(\frac{c}{d})$.

5. Finally, if $\frac{a}{b} < \frac{c}{d}$ with $b, d > 0$, then $ad < bc$. Since j is order-preserving on $\mathbb{Z}$, $j(ad) < j(bc)$, hence $j(a)j(d) < j(b)j(c)$. Because $b, d > 0$, we have $j(b), j(d) > 0$. Dividing by these positive elements preserves the inequality: $\frac{j(a)}{j(b)} < \frac{j(c)}{j(d)}$, so $i(\frac{a}{b}) < i(\frac{c}{d})$.

This completes the proof that i is an injective map that respects the ordered field axioms

$\square$

Since we know that $\mathbb{Q}$ is an ordered field, this result says that $\mathbb{Q}$ is the "minimal" ordered field.

Appendix: The Real Numbers are an Ordered Field

This appendix is concerned with proving that the continuum $\mathbb{R}$ is an ordered field. Addition and multiplication on $\mathbb{R}$ are defined in terms of addition and multiplication on $\mathbb{Q}$, so we will use $\oplus$ and $\otimes$ for addition and multiplication of real numbers, to make sure that there is no confusion with $+$ and $\cdot$ on $\mathbb{Q}$.

Definition 7.30. We define $\oplus$ on $\mathbb{R}$ as follows. Let $A, B \in \mathbb{R}$ be Dedekind cuts. Define

$$A \oplus B = \{a + b \mid a \in A \text{ and } b \in B\}.$$

Exercise 7.31. (a) Prove that $A \oplus B$ is a Dedekind cut.

(b) Prove that $\oplus$ is commutative and associative.

(c) Prove that if $A \in \mathbb{R}$ then $A = \mathbf{0} \oplus A$.

Lemma 7.32. *If $A \in \mathbb{R}$, $r \in \mathbb{Q}$, and $r > 0$, then there exists $s \in A$ such that $s + r \notin A$.*

Lemma 7.33. *Let $A \in \mathbb{R}$. Let*

$$B = \{r \in \mathbb{Q} \mid -r \notin A \text{ but } -r \text{ is not a first point of } \mathbb{Q} \setminus A\}.$$

Then,

(a) $B \in \mathbb{R}$.

(b) $A \oplus B = \mathbf{0}$.

Theorem 7.34. *Every $A \in \mathbb{R}$ has an additive inverse $-A$ such that $A \oplus (-A) = \mathbf{0}$.*

Exercise 7.35. Show that $A < \mathbf{0} \iff \mathbf{0} < -A$.

Exercise 7.36. Show that if $p \in \mathbb{Q}$ and $A = \{x \in \mathbb{Q} \mid x < p\}$, then $-A = \{x \in \mathbb{Q} \mid x < -p\}$.

Exercise 7.37. Show that $<$ satisfies additivity; i.e., if $A, B, C \in \mathbb{R}$ with $A < B$ then $A \oplus C < B \oplus C$.

Definition 7.38. For $A, B \in \mathbb{R}$, $\mathbf{0} < A$, $\mathbf{0} < B$, we define

$$A \otimes B = \{r \in \mathbb{Q} \mid r \leq 0\} \cup \{ab \mid a \in A, b \in B, a > 0, b > 0\}.$$

If $A = \mathbf{0}$ or $B = \mathbf{0}$ we define $A \otimes B = \mathbf{0}$. If $A < \mathbf{0}$ but $\mathbf{0} < B$ we replace A with $-A$ and use the definition of multiplication of positive elements. Hence, in this case,

$$A \otimes B = -[(-A) \otimes B].$$

Similarly, if $\mathbf{0} < A$ but $B < \mathbf{0}$, then

$$A \otimes B = -[A \otimes (-B)],$$

and if $A < \mathbf{0}, B < \mathbf{0}$ then

$$A \otimes B = (-A) \otimes (-B).$$

Exercise 7.39. (a) Show that if $A, B \in \mathbb{R}$, then $A \otimes B \in \mathbb{R}$.

(b) Show that $\otimes$ is commutative and associative.

(c) Show that if $A, B \in \mathbb{R}$, $\mathbf{0} < A$, and $\mathbf{0} < B$, then $\mathbf{0} < A \otimes B$.

(d) Let $\mathbf{1} = \{x \in \mathbb{Q} \mid x < 1\}$. Show that if $A \in \mathbb{R}$, then $\mathbf{1} \otimes A = A$.

Lemma 7.40. *If $A \in \mathbb{R}$, $A > 0$, $r \in \mathbb{Q}$, and $r > 1$, then there exist $s \in A$ such that $rs \notin A$.*
(Hint: Prove that $\{r^n \mid n \in \mathbb{N}\}$ is unbounded in $\mathbb{Q}$.)

Lemma 7.41. *Let $A \in \mathbb{R}$ with $\mathbf{0} < A$. Let*

$$B = \{r \in \mathbb{Q} \mid r \leq 0\} \cup \left\{ r \in \mathbb{Q} \;\middle|\; r > 0 \text{ and } \frac{1}{r} \notin A, \text{ but } \frac{1}{r} \text{ is not a first point of } \mathbb{Q} \setminus A \right\}.$$

Then,

(a) $B \in \mathbb{R}$.

(b) $A \otimes B = \mathbf{1}$.

Lemma 7.42. *Every* $A \in \mathbb{R}$ *with* $\mathbf{0} < A$ *has a multiplicative inverse* A^{-1} *such that* $A \otimes A^{-1} = \mathbf{1}$.

Lemma 7.43. *Let* $A \in \mathbb{R}$ *with* $A < \mathbf{0}$. *Then,* $(-A)^{-1}$ *has already been defined, and* $A \otimes (-(-A)^{-1}) = \mathbf{1}$.

Theorem 7.44. *Every* $A \in \mathbb{R}$ *with* $A \neq \mathbf{0}$ *has a multiplicative inverse* A^{-1} *such that* $A \otimes A^{-1} = \mathbf{1}$.

Exercise 7.45. Show that if $p \in \mathbb{Q}$ with $p \neq 0$ and $A = \{x \in \mathbb{Q} \mid x < p\}$, then $A^{-1} = \{x \in \mathbb{Q} \mid x < \frac{1}{p}\}$.

Theorem 7.46. $\mathbb{R}$ *is an ordered field.*

Additional Exercises

In these exercises you should assume the results from the Appendix, whether or not you proved these in class. Also, the function $i : \mathbb{Q} \to \mathbb{R}$ referred to in exercises 1 and 2 is the function defined in Script 6 - so $i(q) = \{x \in \mathbb{Q} \mid x < q\}$.

1. a) Let $q \in \mathbb{Q}$. Prove that $i(q) \oplus i(-q) = \mathbf{0}$.

   b) Show that for $q_1, q_2 \in \mathbb{Q}$,
$$i(q_1) \otimes i(q_2) = i(q_1 q_2).$$

2. a) Define $\sqrt{2} = \{x \in \mathbb{Q} \mid x < 0\} \cup \{x \in \mathbb{Q} \mid x^2 < 2\}$. We saw in class that $\sqrt{2} \in \mathbb{R}$. Show that $\sqrt{2}$ is not a rational cut, i.e. there is no $p \in \mathbb{Q}$ such that $\sqrt{2} = \{x \in \mathbb{Q} \mid x < p\}$.

   b) Show that $\sqrt{2} \otimes i(x) \in \mathbb{R} \setminus i(\mathbb{Q})$, for $x \in \mathbb{Q}, x > 0$. (Note that in fact this holds for all $x \neq 0$, but you need only show it for $x > 0$.)

   c) Show that the irrationals are dense in the reals - more precisely, show that $\mathbb{R} \setminus i(\mathbb{Q})$ is dense in $\mathbb{R}$.

   d) *(Challenge)* Show that $\sqrt{2} \otimes \sqrt{2} = \{x \in \mathbb{Q} \mid x < 2\}$.

3. Let $A \subset \mathbb{R}$ and assume that $\sup A$ exists. Define $-A := \{x \in \mathbb{R} : -x \in A\}$. Is it true that $-\sup A = \sup -A$?

4. (a) Let A, B be bounded subsets of $\mathbb{R}$. Define $A + B = \{c \in \mathbb{R} : c = a + b \text{ for some } a \in A, b \in B\}$. Is it true that $\sup A + \sup B = \sup(A + B)$?

   (b) Let $f, g : \mathbb{R} \to \mathbb{R}$ be functions. Is it true that $\sup f(\mathbb{R}) + \sup g(\mathbb{R}) = \sup(f + g)(\mathbb{R})$? (Note: here we define $f + g : \mathbb{R} \to \mathbb{R}$ to be the function $(f + g)(c) = f(c) + g(c)$ for all $c \in \mathbb{R}$.)